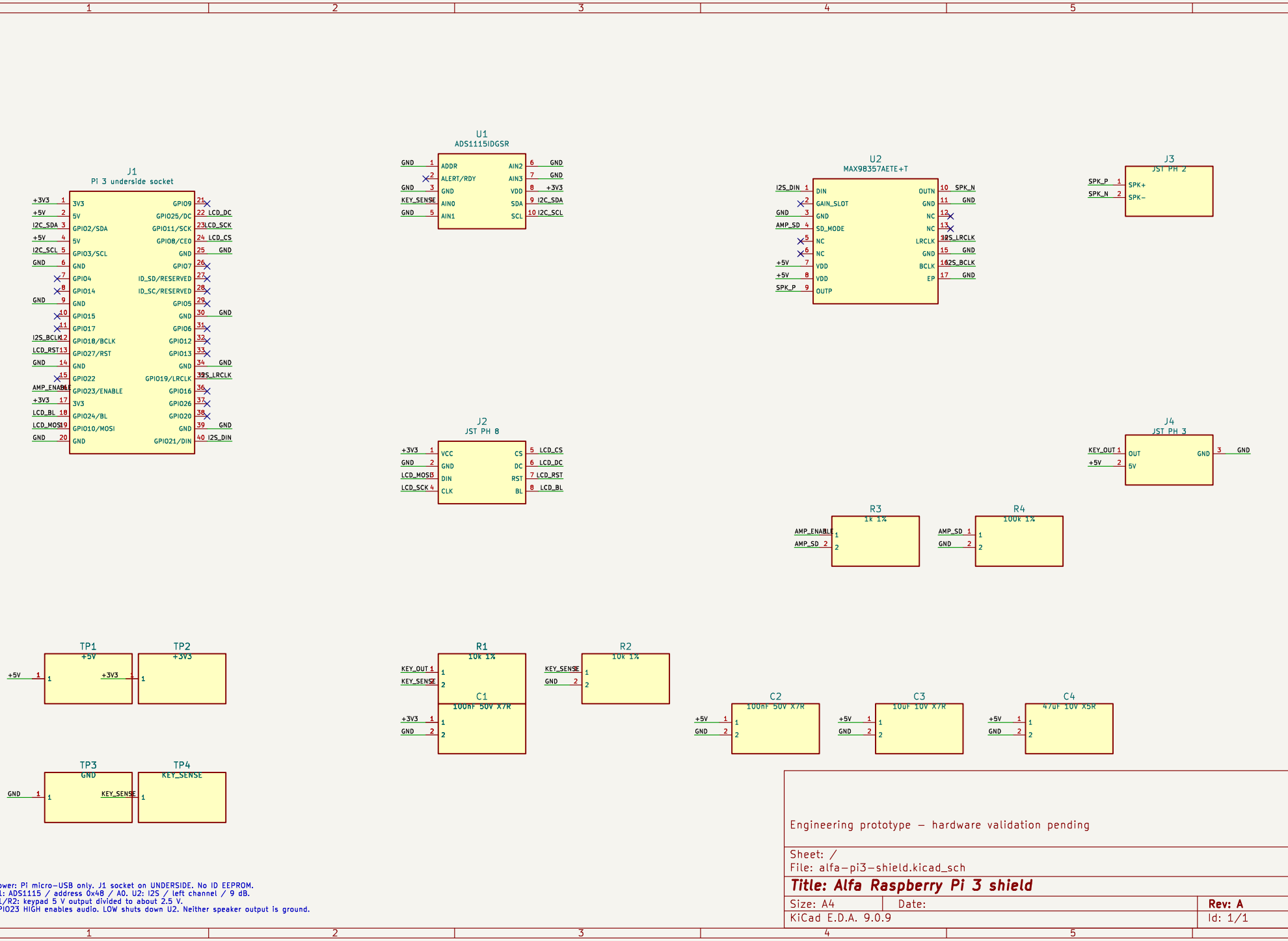

J1
Pi 3 underside socket
3V3
5V
GPIO2/SDA
5V
GPIO3/SCL
GND
GPIO4
GPIO14
GND
GPIO15
GPIO17
GPIO18/BCLK
GPIO27/RST
GND
GPIO22
GPIO23/ENABLE
3V3
GPIO24/BL
GPIO10/MOSI
GND
GPIO9
GPIO25/DC
GPIO11/SCK
GPIO8/CE0
GND
GPIO7
ID_SD/RESERVED
ID_SC/RESERVED
GPIO5
GND
GPIO6
GPIO12
GPIO13
GND
GPIO19/LRCLK
GPIO16
GPIO26
GPIO20
GND
GPIO21/DIN
+3V3
+5V
I2C_SDA
+5V
I2C_SCL
GND
AMP_ENABLE
+3V3
LCD_BL
LCD_MOSI
GND
LCD_DC
LCD_SCK
LCD_CS
GND
I2S_BCLK
LCD_RST
GND
I2S_LRCLK
GND
I2S_DIN
U1
ADS1115IDGSR
ADDR
ALERT/RDY
GND
AIN0
AIN1
AIN2
AIN3
VDD
SDA
SCL
GND
GND
KEY_SENSE
GND
GND
+3V3
I2C_SDA
I2C_SCL
U2
MAX98357AETE+T
DIN
GAIN_SLOT
GND
SD_MODE
NC
NC
VDD
VDD
OUTP
OUTN
GND
NC
NC
LRCLK
GND
BCLK
EP
I2S_DIN
GND
AMP_SD
SPK_P
SPK_N
GND
I2S_LRCLK
GND
I2S_BCLK
GND
J3
JST PH 2
SPK+
SPK-
SPK_P
SPK_N
J2
JST PH 8
VCC
GND
DIN
CLK
CS
DC
RST
BL
+3V3
GND
LCD_MOSI
LCD_SCK
LCD_CS
LCD_DC
LCD_RST
LCD_BL
J4
JST PH 3
OUT
5V
GND
KEY_OUT
+5V
GND
R3
1k 1%
AMP_ENABLE
AMP_SD
R4
100k 1%
AMP_SD
GND
TP1
+5V
+5V
TP2
+3V3
+3V3
TP3
GND
GND
TP4
KEY_SENSE
KEY_SENSE
R1
10k 1%
KEY_OUT
KEY_SENSE
R2
10k 1%
KEY_SENSE
GND
C1
100nF 50V X7R
+3V3
GND
C2
100nF 50V X7R
+5V
GND
C3
10uF 10V X7R
+5V
GND
C4
47uF 10V X5R
+5V
GND
Power: Pi micro-USB only. J1 socket on UNDERSIDE. No ID EEPROM.
U1: ADS1115 / address 0x48 / A0. U2: I2S / left channel / 9 dB.
R1/R2: keypad 5 V output divided to about 2.5 V.
GPIO23 HIGH enables audio. LOW shuts down U2. Neither speaker output is ground.
Engineering prototype — hardware validation pending
Sheet: /
File: alfa-pi3-shield.kicad_sch
Title: Alfa Raspberry Pi 3 shield
Size: A4
Date:
Rev: A
KiCad E.D.A. 9.0.9
Id: 1/1